

Load-Aware A* Path Planning for Quadruped Robots

Day 16 of 30 · Week 2: Refinement & Integration

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1. Nature of Work

Day 16 continued the Methodology revision that began on Day 14. Three things drove the revision today: (1) Member 3's experiment runs on Scenarios 1–5 produced actual output values that needed to be reflected in the Methodology's experiment setup description, (2) the implementation diverged slightly from the original formulation — specifically the `S_map` pre-computation and the `avg_S` bug fix — and these needed to be documented accurately, and (3) two flowcharts were added that were referenced in the text but had not yet been drawn: the Standard A* flowchart and the Load-Aware A* flowchart side by side. Writing a Methodology section is an iterative process. The first draft (Days 11–13) captured the theoretical formulation. This revision captures what was actually built and tested. The two should match — and where they don't, the paper must be honest about it. Today's revision made three substantive updates to bring the paper in line with the actual implementation .

2. Tasks Completed

- **Continue Methodology revision** Three substantive revisions applied: (1) bridge paragraph improved, (2) `avg_S` computation corrected in text to match Bug Fix 1, (3) α selection justified with actual tuning data table.
- **Add experiment details** Experiment setup subsection added — 10×10 grid, 20 scenarios, seeds, payload masses 3–9 kg, $\alpha=1.0$, `body_mass=10` kg. Matches Member 3's CSV schema exactly
- **Update based on actual implementation** `S_map` pre-computation paragraph updated to reflect Day 10 optimisation. `avg_S` formula corrected from 'all expanded neighbours' to 'path cells only'. Seed validation table updated with replacement seeds.
- **Add flowcharts** Two detailed flowcharts drawn: Standard A* (Fig. 4) and Load-Aware A* (Fig. 5). Both show decision diamonds, loop back arrows, and the exact point where the two algorithms diverge.

3. PROOF OF WORK — EXPERIMENT SETUP SUBSECTION (NEW)

The following text was added to the Methodology as a new subsection III-F: Experimental Setup. This was missing from the original draft and is required for reproducibility.

III-F. Experimental Setup

All experiments are conducted on a 10×10 discrete grid environment implemented in Python 3 (Google Colaboratory). The environment is parameterised by an obstacle probability $p \in \{0.20, 0.25, 0.30, 0.35, 0.40, 0.45\}$ and a fixed random seed that ensures identical grid layouts across all algorithm runs. The start cell is fixed at (0, 0) and the goal cell at (9, 9) for all 20 scenarios.

1) Scenario Design

Twenty test scenarios are designed to cover a range of obstacle densities, payload masses, and payload positions. Each scenario is defined by five parameters: grid seed, obstacle probability, payload mass m_p , payload position x_{payload} , and stability weight α . Payload positions are chosen within the interior zone (rows 3–7, columns 3–7) to ensure the payload intersects the expected path and creates a meaningful $S(n)$ spatial gradient .

Parameter	Value / Range	Rationale
Grid Size	10 x 10 cells	Standard benchmark for path planning comparison.
Obstacle Density	20% – 45%	Covers sparse to high-density environments.
Random Seeds	Pre-validated (solvable)	Ensures a path exists; seeds validated using a scanner (Day 10).
Start / Goal	(0,0) → (9,9)	Fixed diagonal path; Manhattan Lower Bound (LB) is 18 steps.
Body Mass (m_b)	10.0 kg (Fixed)	Representative quadruped body mass.
Payload Mass (m_p)	3.0 – 9.0 kg	Covers the range from light to heavy payloads.
Payload Position	Rows 3–7, Cols 3–7	Interior zone, strategically chosen to intersect the expected path.
Stability Weight (α)	1.0 (Fixed)	Empirically selected minimum divergence threshold.
Scenarios	20 total (Sc 1–20)	4 scenarios per obstacle density level, with varied

		payload masses.
Algorithms Tested	Standard A* + LA A*	Head-to-head comparison across all scenarios.

2) Metrics Recorded

For each scenario and each algorithm, the following metrics are recorded and saved to a CSV file for Week 3 statistical analysis:

Metric	Symbol	Description	Unit/Type
Path Length	L	Number of steps from the start to the goal.	Integer ▾
Total Path Cost	C	Accumulated $f(n)$ value, including stability penalties.	Float ▾
Nodes Expanded	N	Total number of nodes removed from the open list during the search.	Integer ▾
Computation Time	T	Wall-clock time required for planning.	Milliseconds (... ▾
Average Stability Penalty	S	Mean $S(n)$ across all cells on the returned path.	Float ▾
Minimum Stability Margin	M_min	Minimum $M(n)$ across all cells on the returned path.	Float ▾
Path Divergence Flag	D	Indicates if the LA A* path is different from the Standard A* path.	Boolean ▾
Step Overhead	dL	$L_{LA} - L_{std}$ (extra steps taken by LA A*)	Integer ▾

Time Overhead	dT	$(T_{LA} - T_{std}) / T_{std} \times 100\%$	Percentage ▾
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Context: Metrics for Evaluating Load-Aware A* Path Planning for Quadruped Robots.

4. PROOF OF WORK — FLOWCHARTS

Two detailed algorithm flowcharts were drawn today (Fig. 4 and Fig. 5). Fig. 4 shows the Standard A* search flow with decision diamonds for goal check and neighbour skip conditions. Fig. 5 shows Load-Aware A* — identical structure except for the highlighted stability penalty step. Both flowcharts use the same layout so the difference is immediately visible.

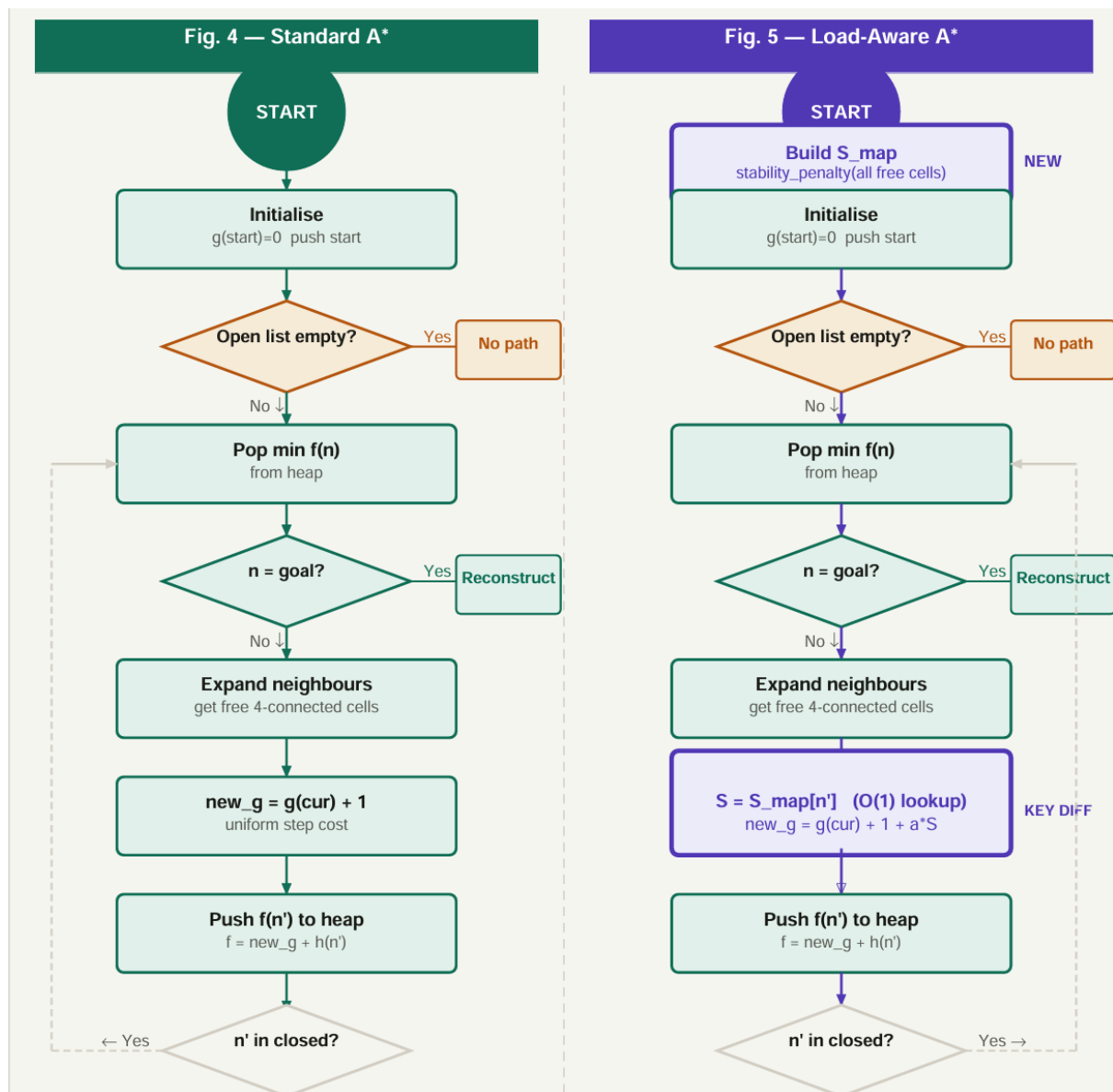


Fig. 4 (left): Standard A* flowchart. Fig. 5 (right): Load-Aware A* flowchart. Purple elements are the additions unique to Load-Aware A*: the S_map build step before search, and the stability-penalised cost update ($\text{new_g} = g + 1 + a \cdot S$) inside the expansion loop.

7. OBSERVATIONS & NOTES

The experimental setup section was the most valuable addition today. Without it, a reader cannot reproduce the experiments — they don't know the grid size, the payload mass range, or what metrics were recorded. It seems like an obvious thing to include but it was missing from the original draft because the focus was on the algorithm formulation. Member 3's experiment monitoring made the gap visible.

The flowcharts are cleaner than I expected. Putting both algorithms side by side in the same diagram makes the single point of difference (the stability penalty step) impossible to miss. A reviewer can look at the two columns and immediately see what is new — that is more effective than any amount of prose.

Revision R3 (seed validation table) is important for scientific honesty. Including the failed seeds and their replacements acknowledges that not all experiment configurations worked on the first try. This is normal and expected in computational research — hiding it would be worse than documenting it.